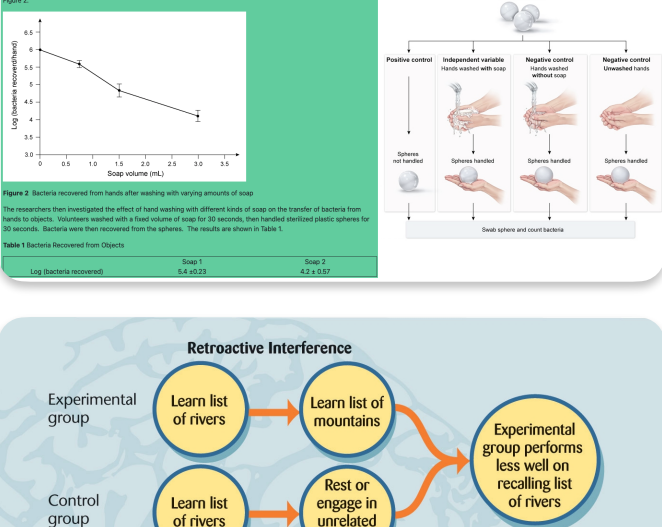
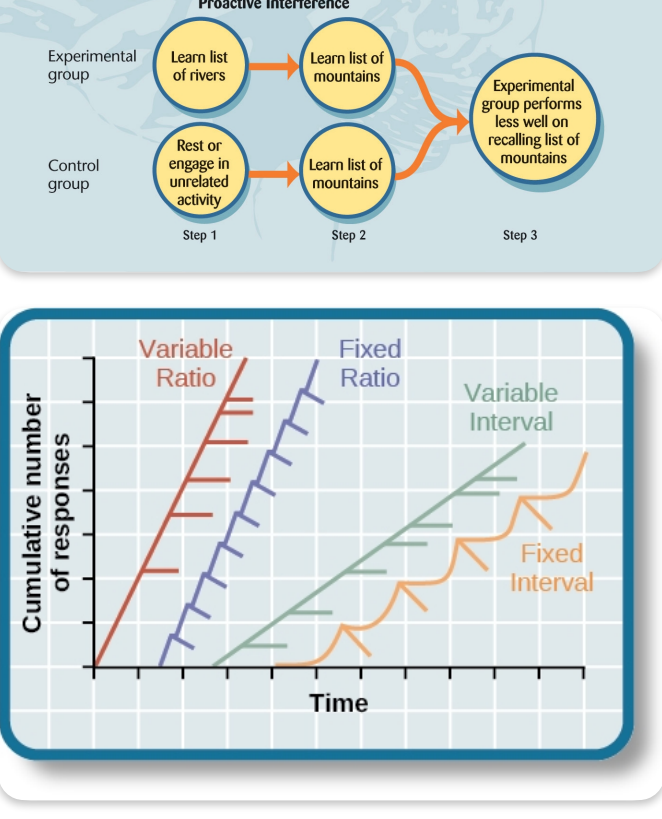




Correlation—The relationship of two variables to each other. In this case, we are looking at how stress and achievement levels are related.

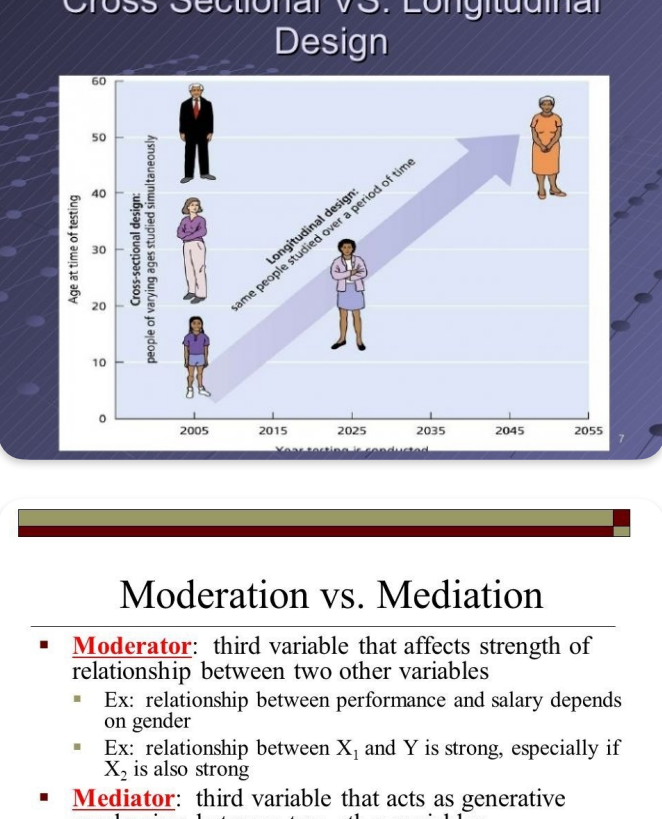
- Negative Correlation**—When two variables move in the *opposite* direction, they are negatively correlated. For example, if high stress levels and low achievement go together and low stress levels and high achievement go together, these variables are *negatively* correlated.
- Positive Correlation**—When two variables move the *same* direction, they are positively correlated. For example, if low SES and low achievement go together and high SES and high achievement go together, these variables are *positively* correlated.

Another concept we should be familiar with is that correlation and causation are not the same thing. Correlation is simply acknowledging that two variables consistently move together (or opposite each other) but it doesn't try to claim that one causes the other (which is causation). Correlation is a much simpler concept to measure and prove than causation.



Prevalence vs. Incidence

- Prevalence:** frequency of existing cases
- Incidence:** frequency of new cases
- New cases are called incident cases.*
- Existing cases are called prevalent cases.*



Classical Twin Study

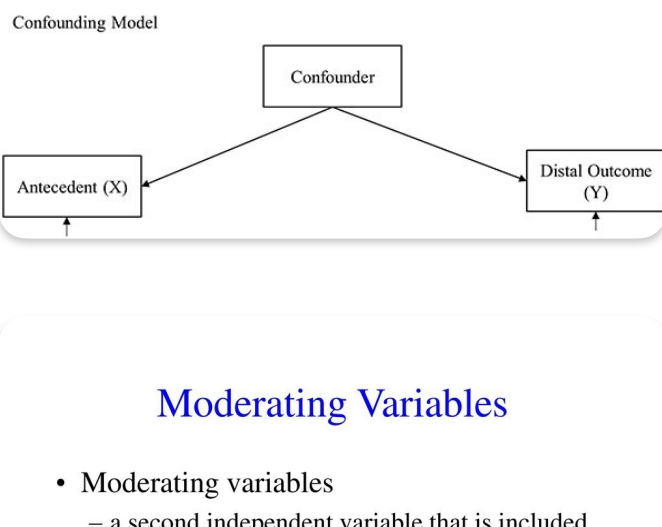
The classical twin study design relies on studying twins raised in the same family environments.

It compares the similarity of identical fraternal twins. Identical twins share all of their genes, while fraternal twins share only about 50 percent of them.

So, if a researcher compares the similarity between sets of identical twins to the similarity between sets of fraternal twins for a particular trait, then any excess likeness between the identical twins should be due to genes rather than environment.

Moderation vs. Mediation

- Moderator:** third variable that affects strength of relationship between two other variables
 - Ex: relationship between performance and salary depends on gender
 - Ex: relationship between X_1 and Y is strong, especially if X_2 is also strong
- Mediator:** third variable that acts as generative mechanism between two other variables
 - Ex: performance mediates relationship between job knowledge and salary
 - Ex: X_1 "causes" X_2 which "causes" Y



	Internal validity	External validity
Describes:	Causality If a change in the independent variable is really causing a change in the dependent variable	Generalizability If the observed relationship applies outside the experiment to another situation or other people
Major threats to:	Confounding variables Selection bias Maturation Repeated testing Regression toward the mean	Artificial research environment Nonrepresentative sample Measurement effects
As experiment becomes more tightly controlled:	↑	↓
As experiment becomes more like the real world:	↓	↑

Generalizability

- A variety of criteria are used to judge the extent to which scientific findings are credible (i.e., can be trusted or believed).
- External validity is concerned with generalizability or the extent to which results can be applied to other situations (e.g., outside the laboratory) or people (e.g., to the population).
- If a study's sample is not representative of the population, its findings will likely not apply to that population because they differ in some way.

Internal validity

- Internal validity is the extent to which a study can be applied to other situations or other people.
- Internal validity is the extent to which a study can be applied to other situations or other people.

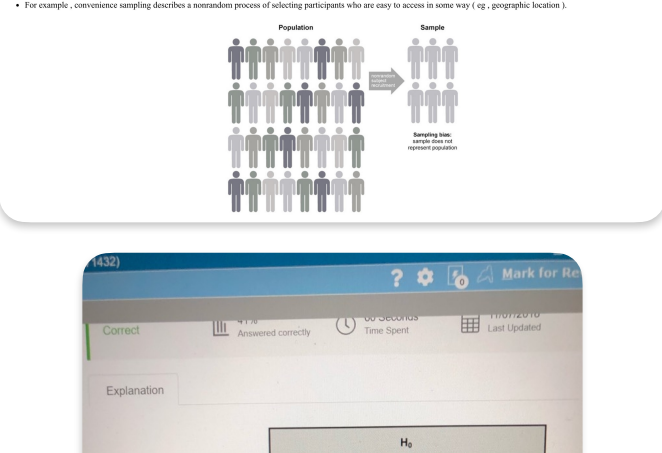
Moderating Variables

- Moderating variables
 - a second independent variable that is included because it is believed to have a significant contributory or contingent effect on the original state IV-DV relationship
- Example
 - The introduction of the four-day workweek (IV) will lead to higher productivity (DV), especially among younger workers (MV)

Convenience Sampling

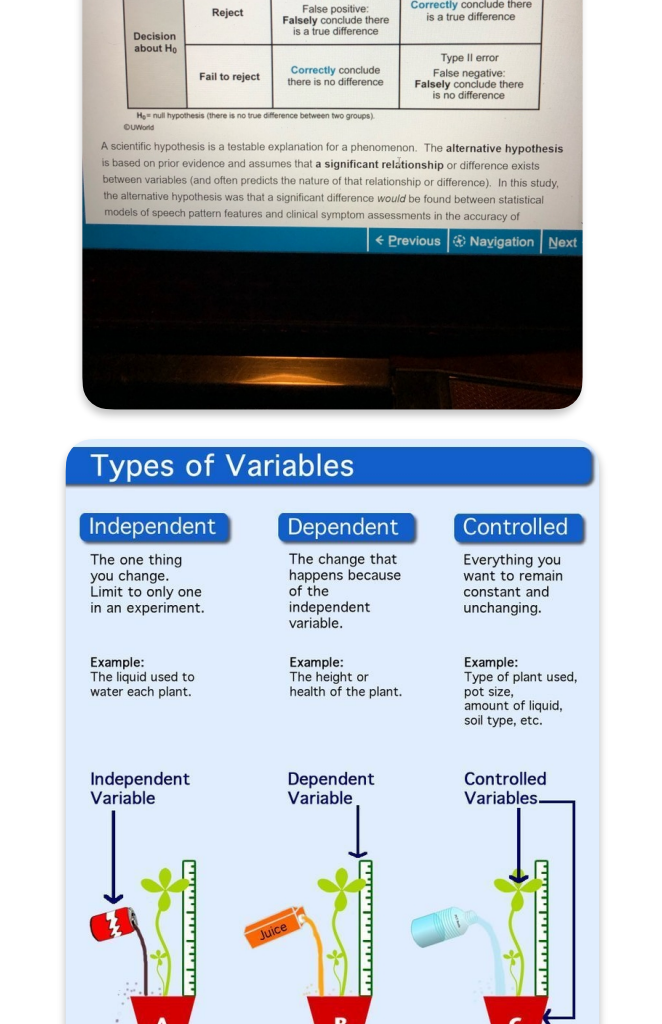
- Describes a nonrandom process of selecting participants who are easy to access in some way (e.g., geographic location).

Type	Description	Example
Convenience sample	The most readily accessible members of a population are included in the study.	Using a sample of Chinese immigrant children to draw conclusions about Chinese children.
Reactive sample	The members who want to participate in the study are included.	Using a sample of children who volunteer to participate to draw conclusions about all children.
Revelant sample	One member of a population participates in the study. A follow-up study is then conducted.	Using a single immigrant child and the results for the study to draw conclusions about all immigrant children.



The Truth (Based on Entire Population)

	Nothing Is There (H_0 Is True)	Something Is There (H_0 Is False)
I Don't See Anything (Nonsignificant)	Right!	Wrong (Type II Error)
I See Something (Significant)	Wrong (Type I Error)	Right!



Now, for **clinical significance**, we need to interpret confidence limits in relation to the **smallest clinically beneficial and harmful effects**.

- These are usually **equal and opposite** in sign.
- They define regions of beneficial, trivial, and harmful values.
- Can use the MCID (or similar measure) if available

harmful (smallest clinically harmful effect) → 0 → beneficial (smallest clinically beneficial effect)

Direction of effect

Confounders: Known & Unknown

- A variable (the confounder) falsely appears to be the cause of the outcome instead of the true cause.

Example: Vitamins (Healthier Lifestyle) → Reduced Risk CHD

Is taking vitamins responsible for reduced CHD risks? Or do people who take vitamins have healthier lifestyles?

Example: Vitamins (Healthier Lifestyle) → Reduced Risk CHD

Outside the study: External Validity
Does the same thing happen in other settings? Other Labs, Everyday settings

Inside the study: Internal Validity
Was the research done "right"?

False Positive vs False Negative

- When diagnosing, detecting, or predicting the development or presence of a medical condition, two types of errors are possible:
 - A false-positive error occurs when the condition is predicted (or said) to be present but in reality is absent.
 - A false-negative error occurs when the condition is predicted (or said) to be absent but in reality is present.

	Condition is present	Condition is absent
Prediction	Correct prediction (condition is present)	False negative (condition predicted to be absent but is present)
	False positive (condition predicted to be present but is absent)	Correct prediction (condition is absent)

Null Hypothesis

- The null hypothesis (H_0) states that there is no difference between 2 groups.
- In essence, the alternative hypothesis (H_A) is the null by claiming that a difference does exist.
- Rejecting a null hypothesis (type I error) leads to the incorrect conclusion that there is a difference between groups (false positive).

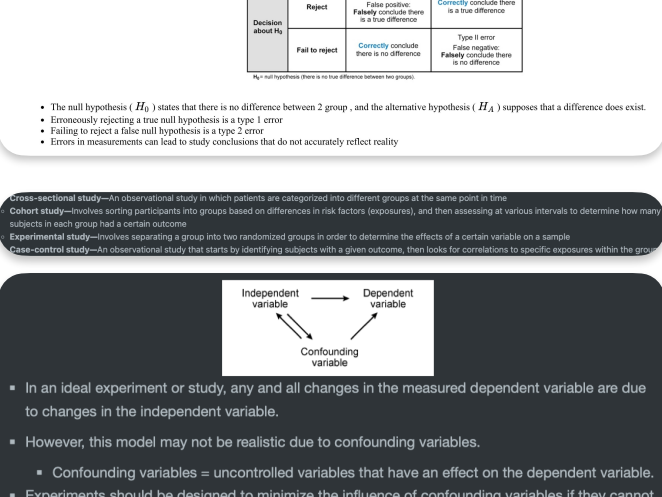
	True	False
Decision about H_0	Reject: Type I error (condition predicted to be present but is absent)	Correctly conclude there is a true difference
	Fail to reject: Correctly conclude there is no difference	Type II error (condition predicted to be absent but is present)

False Positive = "We thought it was there, but it wasn't."

False Negative = "We thought it wasn't there, but it was."

Reject the Null Hypothesis = "Something happened. You said nothing would."

Fail to Reject the Null Hypothesis = "Nothing happened, like you said it would."



Type I Error

You're pregnant!

Type II Error

You're not pregnant!